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(54) FLUSH WATER TANK DEVICE, AND FLUSH TOILET APPARATUS INCLUDING THE SAME

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(57) **ABSTRACT**

The present invention provides a flush water tank device including an outer tank, an inner tank disposed inside the outer tank, a water supply valve, a first water discharge/stop switching mechanism disposed inside the outer tank and outside the inner tank to switch supply and stop of flush water stored in the outer tank to the flush toilet main body, and a second water discharge/stop switching mechanism disposed inside the inner tank to switch supply and stop of flush water stored in the inner tank to the flush toilet main body, the water supply valve is configured to supply flush water into the inner tank, and the flush water supplied into the inner tank overflows from two or more sides of the inner tank to flow into the outer tank.

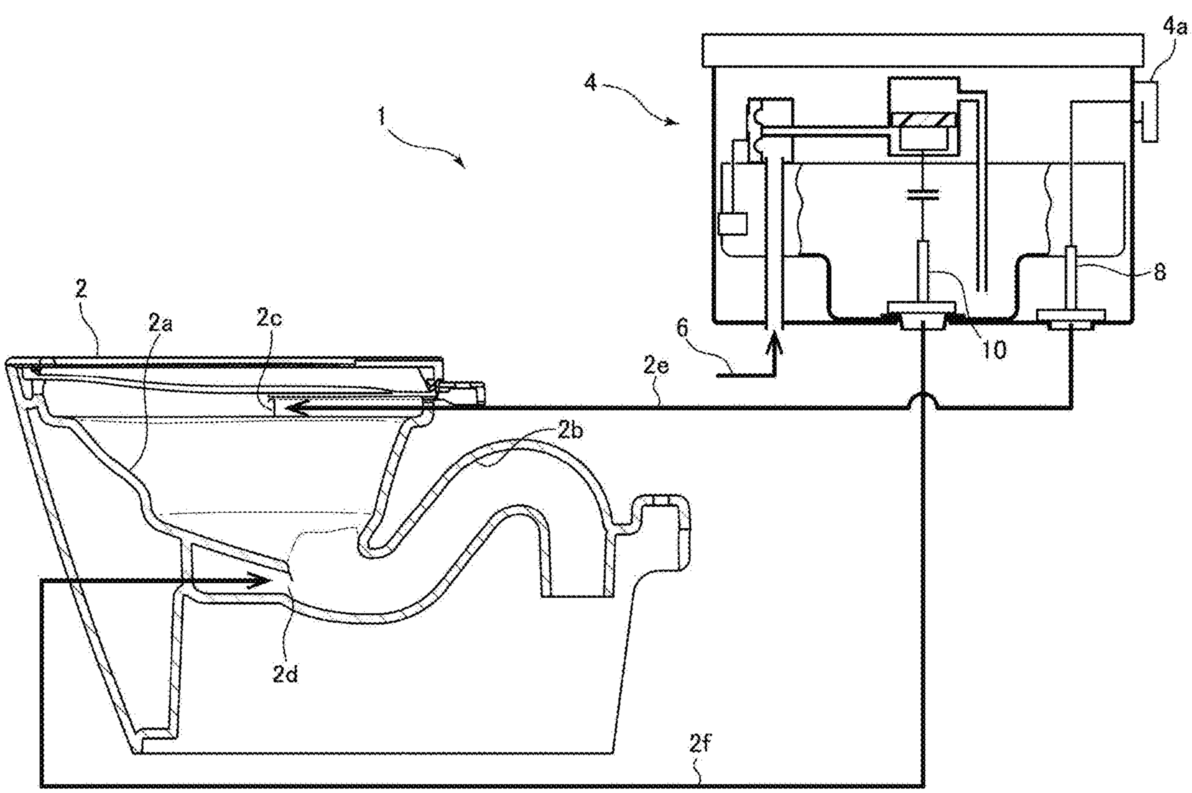


FIG.1

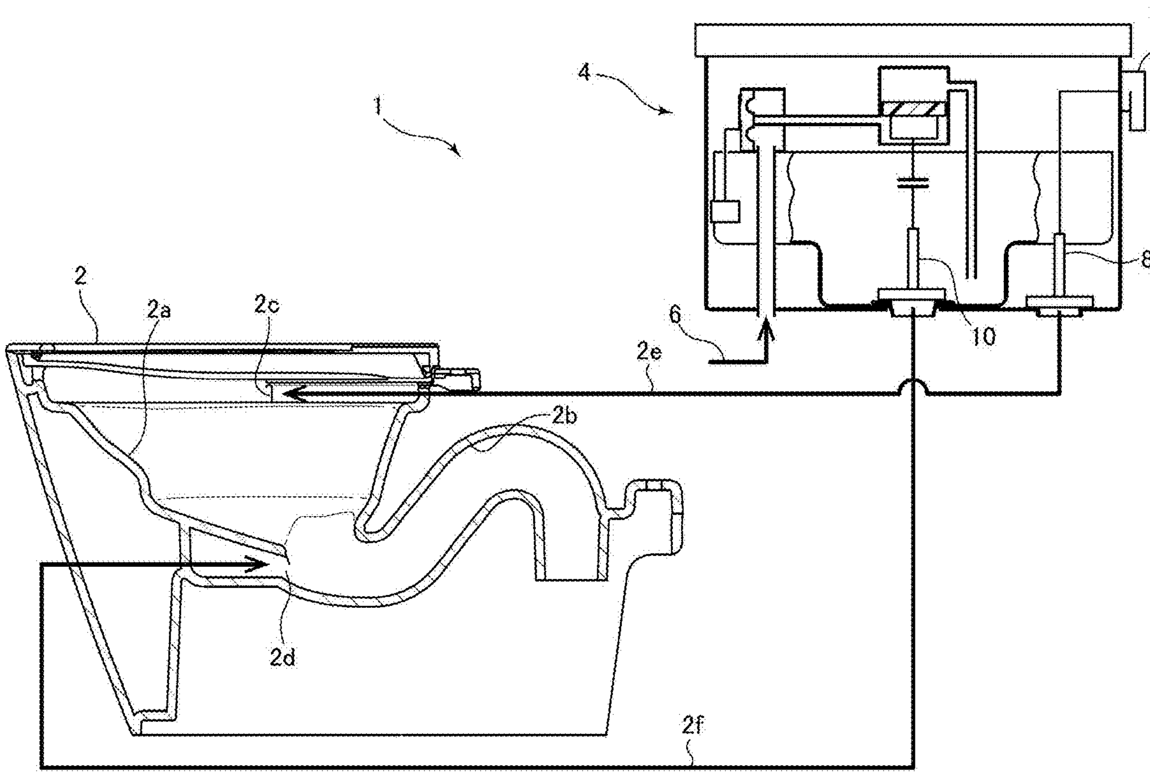


FIG.2

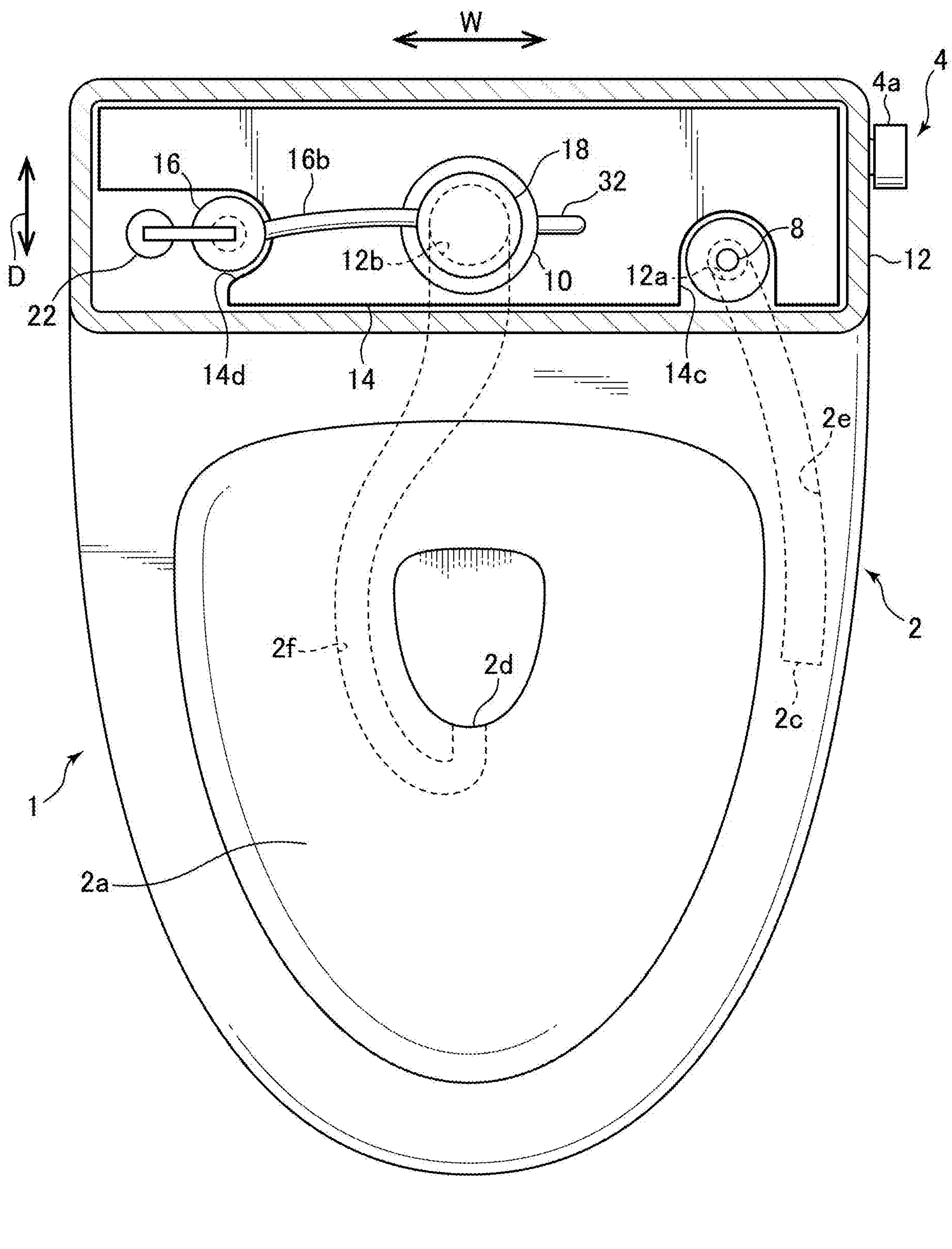
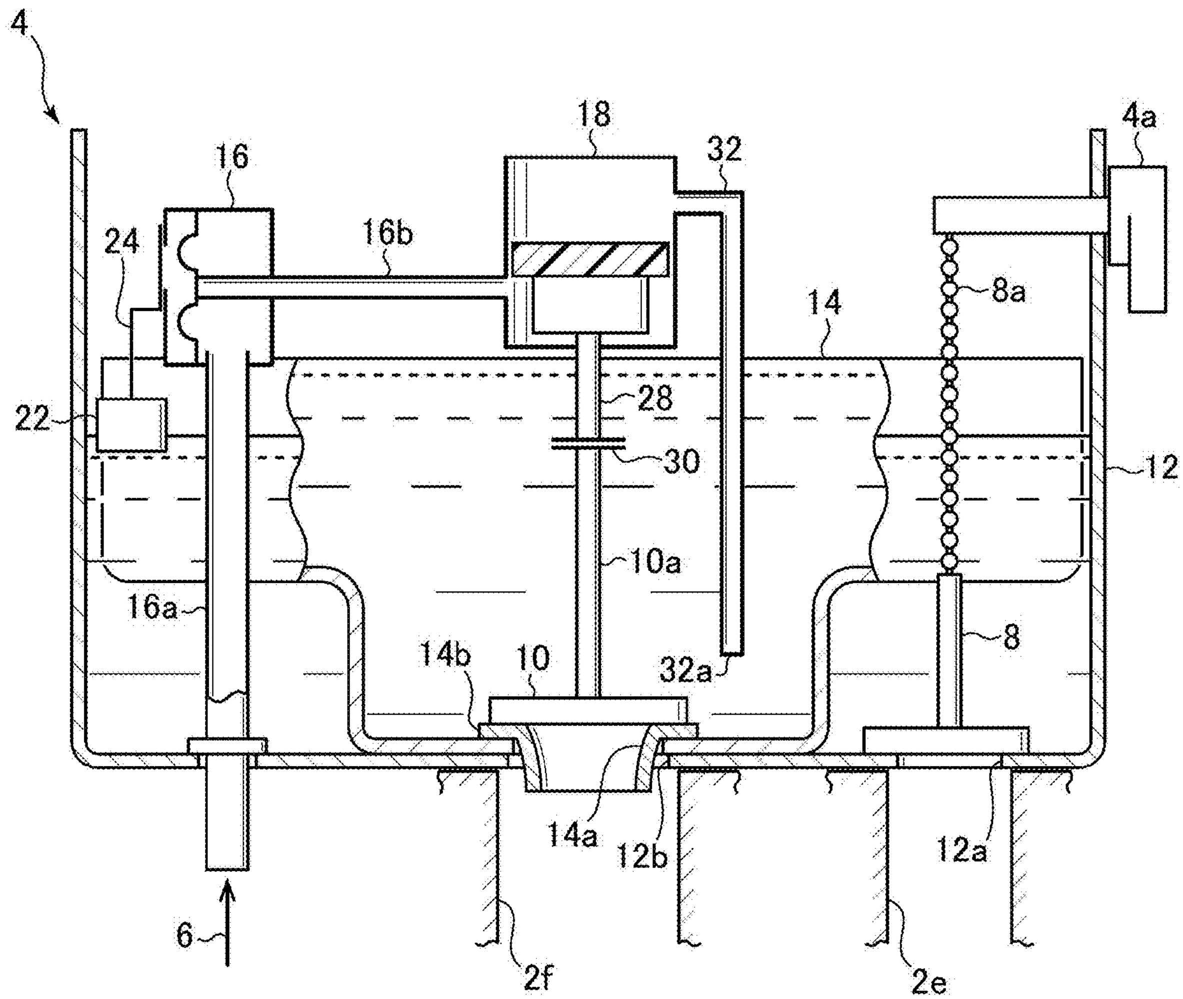


FIG.3


4
18
32
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14a
12b
12a
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FIG.4

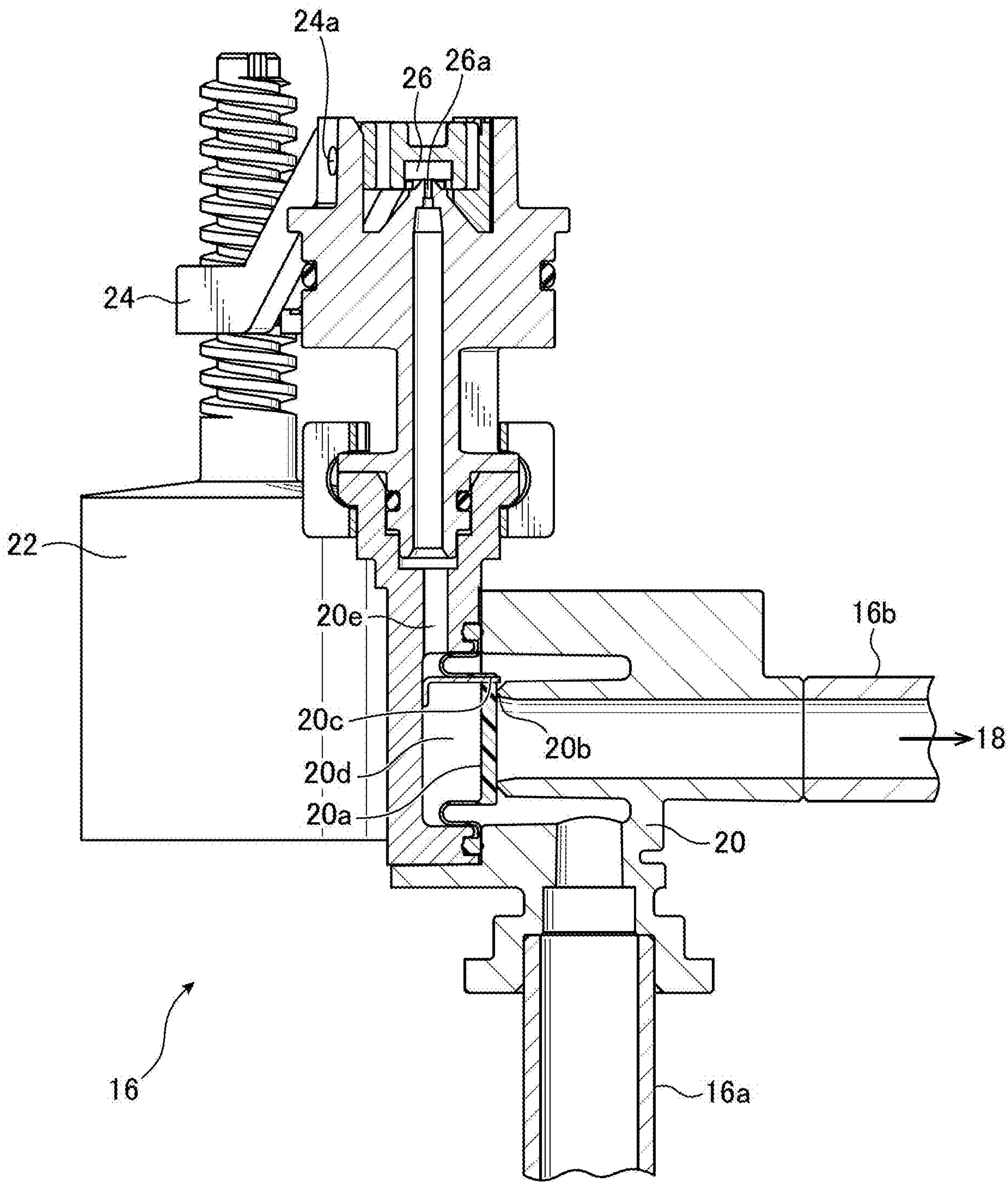


FIG.5

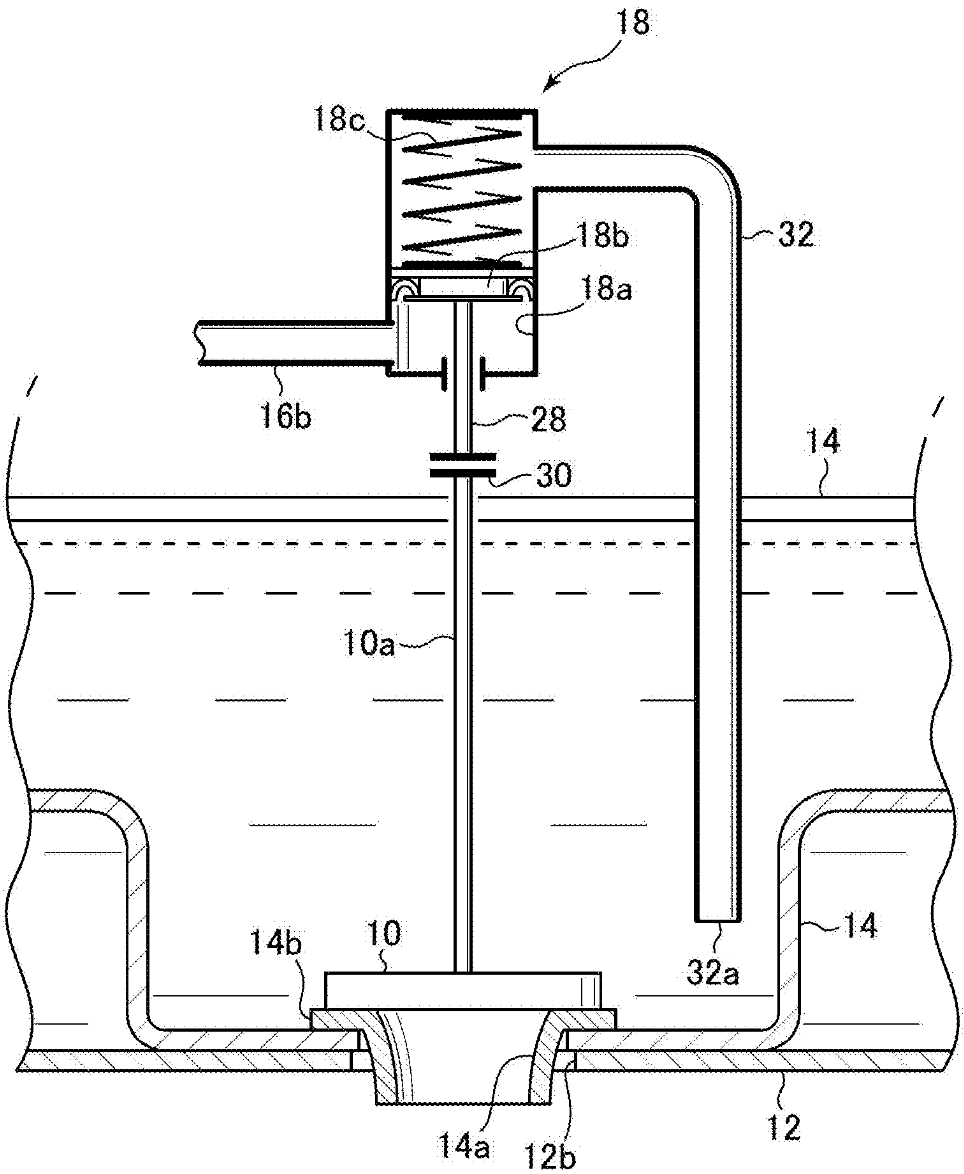
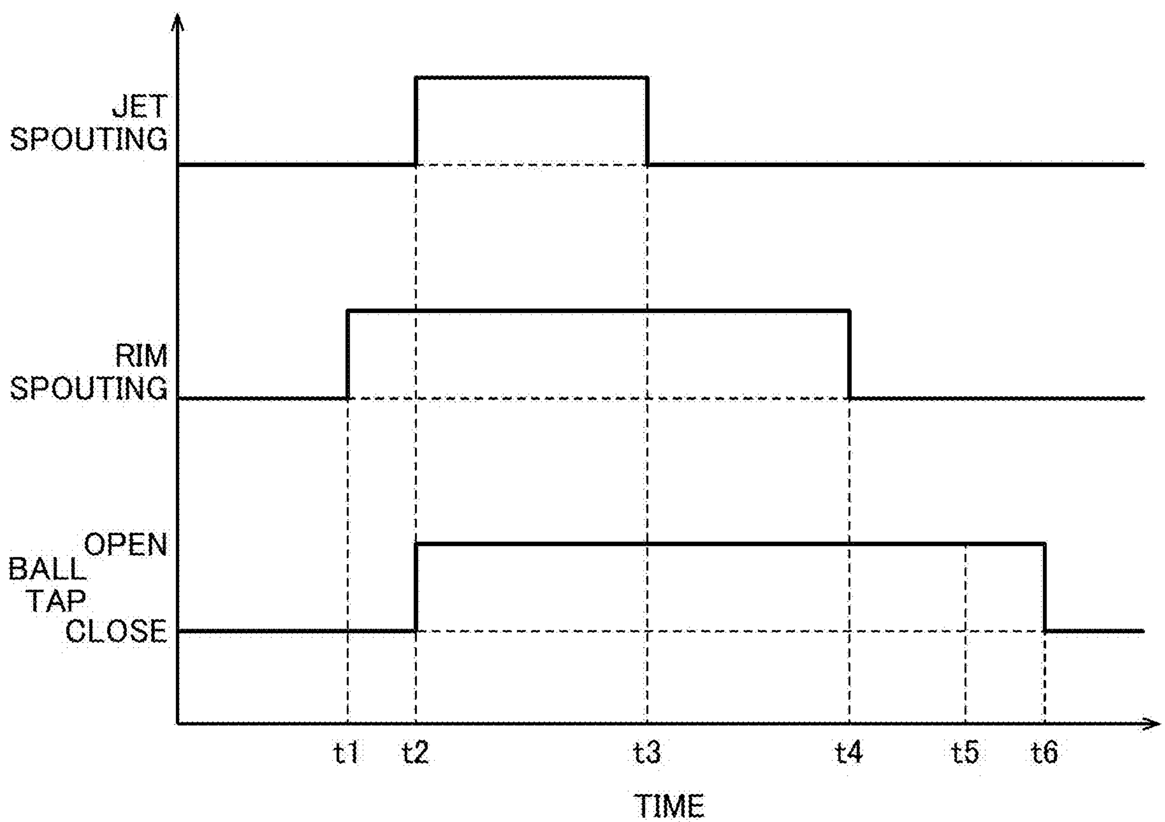


FIG.6


JET SPOUTING
RIM SPOUTING
BALL TAP
OPEN
CLOSE
t1
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t6
TIME

FLUSH WATER TANK DEVICE, AND FLUSH TOILET APPARATUS INCLUDING THE SAME

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims the benefit of Japanese Patent Application No. 2024-029550 filed on Feb. 29, 2024, which is incorporated by reference herein in its entirety.

BACKGROUND OF THE INVENTION

Field of the Invention

[0002] The present invention relates to a flush water tank device, particularly to a flush water tank device attached to a flush toilet main body for use to supply flush water, and a flush toilet apparatus including the flush water tank device.

Description of the Related Art

[0003] In Japanese Utility Model Laid-Open No. 6-056183 (Patent Literature 1), a toilet flushing water supply device is described. In the toilet flushing water supply device, a tank for storing flush water is divided into two by a partition plate, and a discharge valve is provided in each of the divided tanks. In addition, on one side of the tank, which is separated by the partition plate, a ball tap faucet is provided, and flush water supplied from a water supply source flows in. If a water level in the tank on the side on which the ball tap faucet is provided rises beyond the height of the partition plate, flush water flows over a top edge of the partition plate and into a side on which the ball tap faucet is not provided. In the toilet flushing water supply device described in Patent Literature 1, flush water is supplied to both the tanks separated by the partition plate through one water supply valve (ball tap faucet).

[0004] As in the toilet flushing water supply device described in Patent Literature 1, a configuration in which the discharge valve is provided in each of divided flush water tanks is capable of accurately setting an amount of flush water supplied via each discharge valve and is therefore effective in saving water and improving flushing performance. In the toilet flushing water supply device described in Patent Literature 1, however, there is a problem that it is difficult to design the flush water tank with a low profile because of a structure in which flush water supplied from the water supply valve flows over the partition plate and flows from one of the tanks into the other tank. That is, in general, various mechanisms and electrical components, such as a device for driving the discharge valve, are arranged above the water surface of the flush water tank and cause a failure if covered with water. Considering that flush water may flow over the partition plate, however, it is necessary to secure a predetermined space above the height of the overflowing flush water before arranging the electrical components and the like. Consequently, in the toilet flushing water supply device described in Patent Literature 1, it is necessary to secure a large space above the top edge of the partition plate, and this makes it difficult to design the flush water tank with the low profile.

[0005] Therefore, an object of the present invention is to provide a flush water tank device that can be designed with a low profile while including a plurality of tanks for storing flush water, and a flush toilet apparatus including the flush water tank device.

SUMMARY OF THE INVENTION

[0006] To solve the above-described problems, the present invention provides a flush water tank device which is attached to a flush toilet main body for use to supply flush water, including an outer tank for storing flush water to be supplied to the flush toilet main body, an inner tank made of resin, which is disposed inside the outer tank and stores flush water to be supplied to the flush toilet main body, a water supply valve that switches supply and stop of flush water to be stored in the outer tank and the inner tank, a first water discharge/stop switching mechanism disposed inside the outer tank and outside the inner tank to switch supply and stop of flush water stored in the outer tank to the flush toilet main body, and a second water discharge/stop switching mechanism disposed inside the inner tank to switch supply and stop of flush water stored in the inner tank to the flush toilet main body, wherein the water supply valve is configured to supply flush water into the inner tank, and the flush water supplied into the inner tank overflows from two or more sides of the inner tank to flow into the outer tank.

[0007] According to the present invention including this configuration, the water supply valve is configured to supply flush water into the inner tank, and the flush water supplied into the inner tank overflows from the two or more sides of the inner tank to flow into the outer tank. Therefore, an edge over which the flush water flows can be lengthened, and a water level of flush water flowing over the edge of the inner tank can be kept low. Consequently, a space to be provided above the inner tank can be set low, and the flush water tank device can be designed with a low profile.

[0008] In the present invention, preferably, the inner tank is configured so that a length of an outer perimeter of an upper end of the inner tank is ½ or more of a length of an outer perimeter of an upper end of the outer tank.

[0009] According to the present invention including this configuration, since the length of the outer perimeter of the upper end of the inner tank is set to ½ or more of the length of the outer perimeter of the upper end of the outer tank, an edge of the inner tank over which the flush water flows can be set sufficiently long, and the water level of the flush water flowing over the edge of the inner tank can be kept low.

[0010] In the present invention, preferably, the inner tank is configured so that a cross-sectional area varies in a height direction and so that a cross-sectional area of an upper part of the inner tank is larger than a cross-sectional area of a lower part.

[0011] According to the present invention including this configuration, since the inner tank is configured so that the cross-sectional area of the upper part of the inner tank is larger than the cross-sectional area of the lower part, the edge of the inner tank over which the flush water flows can be formed long. As a result, the water level of the flush water flowing over the edge of the inner tank can be kept low. Furthermore, since the inner tank is configured so that the cross-sectional area of the upper part is larger than the cross-sectional area of the lower part, a large amount of flush water having a large head pressure is stored in the inner tank. Consequently, an instantaneous flow rate of flush water drained from the inner tank can be increased, and the flush toilet main body can be effectively flushed.

[0012] In the present invention, preferably, the water supply valve is configured to cause flush water to flow into the inner tank from a water supply port provided below the water surface of the inner tank.

[0013] According to the present invention including this configuration, since the water supply valve is configured to cause flush water to flow into the inner tank from the water supply port provided below the water surface of the inner tank, rippling of the flush water flowing into the inner tank from the water supply port can be suppressed. Consequently, even if the space above the inner tank is set low, concern of equipment disposed above the inner tank being covered with water can be reduced, and the flush water tank device can be designed with a low profile.

[0014] In the present invention, preferably, the inner tank is configured so that a length of an outer perimeter of an upper end of the inner tank is longer than a length of an outer perimeter of an upper end of the outer tank.

[0015] According to the present invention including this configuration, since the inner tank is configured so that the length of the outer perimeter of the upper end of the inner tank is longer than the length of the outer perimeter of the upper end of the outer tank, the edge of the inner tank over which flow water flows can be set extremely long, and the water level of the flush water flowing over the edge of the inner tank can be kept sufficiently low.

[0016] Furthermore, the present invention provides a flush toilet apparatus which is flushed with flush water stored in a flush water tank, and the flush toilet apparatus includes a flush toilet main body including a bowl, a rim spout port, and a jet spout port, and the flush water tank device of the present invention that supplies flush water to the rim spout port and the jet spout port of the flush toilet main body.

[0017] According to a flush water tank device of the present invention and a flush toilet apparatus including the same, a low profile can be achieved while providing a plurality of tanks for storing flush water.

BRIEF DESCRIPTION OF THE DRAWINGS

[0018] FIG. **1** is a block diagram showing an entire configuration of a flush toilet apparatus according to an embodiment of the present invention;

[0019] FIG. **2** is a top view showing a schematic configuration of the flush toilet apparatus according to the embodiment of the present invention;

[0020] FIG. **3** is a front cross-sectional view showing a schematic configuration of a flush water tank device provided in the flush toilet apparatus according to the embodiment of the present invention;

[0021] FIG. **4** is a cross-sectional view showing a structure of a ball tap built in the flush water tank device according to the embodiment of the present invention;

[0022] FIG. **5** is a cross-sectional view showing a structure of a hydraulic drive mechanism built in the flush water tank device according to the embodiment of the present invention; and

[0023] FIG. **6** is a time chart showing the operation of a flush toilet apparatus according to the embodiment of the present invention.

DETAILED DESCRIPTION OF THE INVENTION

[0024] Next, with reference to the accompanying drawings, a flush water tank device according to an embodiment of the present invention and a flush toilet apparatus including the same will be described.

[0025] FIG. **1** is a block diagram showing an entire configuration of the flush toilet apparatus according to the embodiment of the present invention. FIG. **2** is a top view showing a schematic configuration of the flush toilet apparatus according to the embodiment of the present invention. FIG. **3** is a front cross-sectional view showing a schematic configuration of a flush water tank device provided in the flush toilet apparatus according to the embodiment of the present invention. FIG. **4** is a cross-sectional view showing a structure of a ball tap built in the flush water tank device according to the embodiment of the present invention. FIG. **5** is a cross-sectional view showing a structure of a hydraulic drive mechanism built in the flush water tank device according to the embodiment of the present invention.

[0026] As shown in FIGS. **1** and **2**, a flush toilet apparatus **1** according to the embodiment of the present invention includes a flush toilet main body **2** and a flush water tank device **4** disposed at the rear of the flush toilet main body **2**. The flush toilet apparatus **1** of the present embodiment is configured to be flushed by operating a lever handle **4*a*** provided in the flush water tank device **4** after use.

[0027] The flush toilet main body **2** includes a bowl **2*a*** and a discharge trap conduit **2*b*** extending from a lower part of the bowl **2*a***. Furthermore, a rim spout port **2*c*** is provided in a top edge portion of the bowl **2*a***, and a jet spout port **2*d*** is provided in the lower part of the bowl **2*a***. During toilet flushing, flush water is spouted from each of the rim spout port **2*c*** and the jet spout port **2*d*** at a predetermined timing, and a waste receiving surface of the bowl **2*a*** is flushed, while waste and flush water in the bowl **2*a*** are drained to the discharge trap conduit **2*b***. The waste and flush water drained to the discharge trap conduit **2*b*** are drained through a discharge socket (not shown) to a sewage pipe (not shown).

[0028] Flush water is supplied to the flush water tank device **4** from a water supply source **6** such as a water supply, and the supplied flush water is stored in the flush water tank device **4**. Furthermore, in the flush water tank device **4**, a first discharge valve **8** that is a first water discharge/stop switching mechanism, and a second discharge valve **10** that is a second water discharge/stop switching mechanism are built and are configured to open and close drain ports, respectively, provided in a bottom portion of the flush water tank device **4**.

[0029] In the present embodiment, by opening the first discharge valve **8**, flush water is spouted from the rim spout port **2*c*** through a rim conduit **2*e*** formed inside the flush toilet main body **2**. Furthermore, by opening the second discharge valve **10**, flush water is spouted from the jet spout port **2*d*** through a jet conduit **2*f*** formed inside the flush toilet main body **2**.

[0030] Next, with reference to FIGS. **2** and **3**, an internal structure of the flush water tank device **4** will be described.

[0031] As shown in FIGS. **2** and **3**, the flush water tank device **4** includes an outer tank **12**, an inner tank **14** disposed inside the outer tank, the first discharge valve **8** disposed in the outer tank **12**, the second discharge valve **10** disposed in the inner tank **14**, a ball tap **16** that is a water supply valve, and a hydraulic drive mechanism **18**.

[0032] The outer tank **12** and the inner tank **14** are containers configured to store flush water to be supplied to the flush toilet main body **2**. In the present embodiment, the outer tank **12** is made of ceramic, and the inner tank **14** disposed inside the outer tank **12** is made of resin.

[0033] Furthermore, as shown in FIG. **3**, a first drain port **12***a* and a second drain port **12***b* are provided on a bottom surface of the outer tank **12**, and in the present embodiment, these drain ports are formed in a circular shape. Here, the flush water stored inside the outer tank **12** (and outside the inner tank **14**) flows through the first drain port **12***a* into the rim conduit **2***e* of the flush toilet main body **2**, and the flush water stored inside the inner tank **14** flows through the second drain port **12***b* into the jet conduit **2***f* of the flush toilet main body **2**.

[0034] The inner tank **14** is disposed on the bottom surface of the outer tank **12**, and the lower part of the inner tank **14** is submerged in the flush water stored in the outer tank **12**. Furthermore, a circular drain port **14***a* is formed on the bottom surface of the inner tank **14**. The drain port **14***a* of the inner tank **14** is disposed concentrically to match the second drain port **12***b* provided in the outer tank **12**. That is, in the present embodiment, the center of the circular second drain port **12***b* and the circular drain port **14***a* coincide in top view. Therefore, the flush water in the inner tank **14** flows into the jet conduit **2***f* of the flush toilet main body **2** through the drain port **14***a* of the inner tank **14** and the second drain port **12***b* of the outer tank **12**.

[0035] Furthermore, as shown in FIG. **3**, in the present embodiment, the drain port **14***a* of the inner tank **14** is composed of a drain port forming member **14***b* composed separately from a main body part of the inner tank **14**. The drain port forming member **14***b* is a tubular member and is water-tightly attached to the bottom surface of the inner tank **14** to form the drain port **14***a* inside. Furthermore, a seat surface is provided at an upper end of the drain port forming member **14***b*, and the drain port **14***a* is closed when the second discharge valve **10** is seated on the seat surface. Therefore, in the present embodiment, the seat surface on which the second discharge valve **10** is seated is composed of a member separate from the inner tank **14**.

[0036] Furthermore, as shown in FIG. **2**, the outer tank **12** of the flush water tank device **4** is formed in a substantially rectangular shape that is long in a width direction of the flush toilet main body **2** (arrow W direction in FIG. **2**) and short in a depth direction (arrow D direction in FIG. **2**) in top view. Furthermore, the inner tank **14** disposed inside the outer tank **12** is also formed long in the width direction and short in the depth direction in top view and is irregularly shaped to avoid the first discharge valve **8** and the ball tap **16**.

[0037] Specifically, in the present embodiment, the ball tap **16** is disposed in a front left end portion (lower left corner in FIG. **2**) in the outer tank **12**, and the first discharge valve **8** is disposed in a front right end portion (lower right corner in FIG. **2**). The inner tank **14** is provided with cutouts **14***c* and **14***d* in opposite front end portions in the width direction in top view to avoid the ball tap **16** and the first discharge valve **8**.

[0038] That is, the inner tank **14** includes an upper part having about the same outer shape as the outer tank **12** in top view. The cutout **14***c* that is U-shaped is then provided in the inner tank **14** in a portion in which the first discharge valve **8** is disposed, and the cutout **14***d* is provided in a portion in which the ball tap **16** is disposed, so that the valve and ball tap do not interfere with the inner tank **14**.

[0039] Furthermore, in the present embodiment, the inner tank **14** is configured to have an upper end at the same height along an entire perimeter. Consequently, when the flush water supplied into the inner tank **14** overflows to flow into the outer tank **12**, flush water flowing over the entire perimeter of the upper end edge of the inner tank **14** flows into the outer tank. Consequently, an edge over which the flush water in the inner tank **14** flows can have a length sufficiently increased. Preferably, the inner tank **14** is configured so that the flush water in the inner tank **14** overflows from two or more sides of the inner tank **14** to flow into the outer tank **12**.

[0040] Furthermore, as shown in FIG. **2**, in the present embodiment, an outer peripheral wall of the inner tank **14** is formed in a shape substantially identical to a shape of an inner peripheral wall of the outer tank **12**, and a small gap is formed between the outer peripheral wall of the inner tank **14** and the inner peripheral wall of the outer tank **12**. As a result, in the present embodiment, a length of an outer perimeter of the upper end of the inner tank **14** is longer than a length of an outer perimeter of an upper end of the outer tank **12** (the length of the outer perimeter is increased by providing the cutouts **14***c* and **14***d* in the inner tank **14**). Consequently, it is possible to extremely increase the length of an edge over which flush water flows when the flush water supplied into the inner tank **14** overflows to flow into the outer tank. Preferably, the length of the outer perimeter of the upper end of the inner tank **14** is set to ½ or more of the length of the outer perimeter of the upper end of the outer tank **12**.

[0041] On the other hand, as shown in FIG. **3**, a stepped portion is formed on each side of the inner tank **14** in the width direction, and a lower part of the inner tank is configured to have a reduced width. That is, the inner tank **14** is substantially formed in a T-shape in front view so that at least the lower part of the inner tank **14** is submerged in flush water stored in the outer tank **12**. In other words, the inner tank **14** is configured so that a cross-sectional area of the inner tank varies in a height direction and so that a horizontal cross-sectional area in the upper part of the inner tank **14** is larger than a horizontal cross-sectional area in the lower part. The upper part of the inner tank **14** is formed large, so that flush water having high potential energy is stored in the inner tank **14**. Consequently, the hydraulic head pressure of the flush water drained from the inner tank **14** can be increased, and an instantaneous flow rate of the drained flush water can be increased.

[0042] Thus, by irregularly shaping the inner tank **14** with resin, sufficient volume is acquired in the inner tank **14** while suppressing a dimension of the flush water tank device **4** in the depth direction. Additionally, the inner tank **14** made of resin can be configured with a thinner wall surface than an inner tank made of ceramic, and hence the volume in the outer tank **12** can be effectively utilized.

[0043] Next, the first discharge valve **8** that is the first water discharge/stop switching mechanism is a valve body disposed to open and close the first drain port **12***a* provided in the outer tank **12**, and the first drain port **12***a* is opened by pulling the first discharge valve **8** upward. Consequently, the flush water in the outer tank **12** is drained to the rim conduit **2***e* (FIG. **1**) of the flush toilet main body **2** and spouted from the rim spout port **2***c*. Therefore, the first discharge valve **8**, which is the first water discharge/stop switching mechanism, switches the supply and stop of the flush water stored in the outer tank **12** to the flush toilet main body **2**.

[0044] In the present embodiment, a bead chain **8***a* (FIG. **3**) coupled to the first discharge valve **8** is pulled by a user rotating and operating the lever handle **4***a* provided in the flush water tank device **4**, and the first discharge valve **8** is pulled upward. As a modification, the present invention may include a configuration in which the first discharge valve **8** is pulled up based on a control signal from a remote controller (not shown) or a detection signal from a human sensor (not shown), to perform flushing.

[0045] Furthermore, in the present embodiment, the first discharge valve **8** is provided as the first water discharge/stop switching mechanism, and a configuration other than the first discharge valve **8** may be used as the first water discharge/stop switching mechanism. For example, a jet pump (not shown) that causes flush water stored in the outer tank **12** to flow into the flush toilet main body may be used as the first water discharge/stop switching mechanism. In this case, a jet pump nozzle (not shown) is disposed inside the outer tank **12**, and through this nozzle, part of the flush water supplied via a water supply valve such as the ball tap **16** is injected. By injecting flush water from the nozzle in the outer tank **12**, the flush water stored in the outer tank **12** can be entrained, and the flush water in the outer tank **12** can flow into the flush toilet main body by jet pump action. In addition to this jet pump (not shown), an arbitrary configuration that switches supply and stop of flush water to the flush toilet main body **2** may be used as the first water discharge/stop switching mechanism.

[0046] Next, the second discharge valve **10** that is a second water discharge/stop switching mechanism is a valve body disposed to open and close the drain port **14***a* provided in the inner tank **14**, and by pulling the second discharge valve **10** upward, the second discharge valve **10** is unseated from the seat surface of the drain port **14***a*, to open the drain port **14***a*. Consequently, the flush water in the inner tank **14** is drained from the drain port **14***a*, flows through the second drain port **12***b* into the jet conduit **2***f* (FIG. **1**) of the flush toilet main body **2**, and is spouted from the jet spout port **2***d*. Therefore, the second discharge valve **10**, which is the second water discharge/stop switching mechanism, switches the supply and stop of the flush water stored in the inner tank **14** to the flush toilet main body **2**.

[0047] In the present embodiment, the second discharge valve **10** is configured to be pulled up from the drain port **14***a* by the hydraulic drive mechanism **18**. That is, the second discharge valve **10** is a valve body including a valve shaft **10***a* extending upward, and the valve shaft **10***a* is pulled up by the hydraulic drive mechanism **18**. Then, when pulled up to a predetermined height, the second discharge valve **10** is disconnected from the hydraulic drive mechanism **18** and gently descends to close the drain port **14***a*. The configuration of the hydraulic drive mechanism **18** will be described later, and the present invention may be applied to a flush water tank device that is not provided with the hydraulic drive mechanism **18**.

[0048] Furthermore, in the present embodiment, the second discharge valve **10** is provided as the second water discharge/stop switching mechanism, and a configuration other than the second discharge valve **10** may be used as the second water discharge/stop switching mechanism. For example, a jet pump (not shown) that causes flush water stored in the inner tank **14** to flow into the flush toilet main body may be used as the second water discharge/stop switching mechanism. In this case, a jet pump nozzle (not shown) is disposed in the inner tank **14**, and through this nozzle, part of the flush water supplied via a water supply valve such as the ball tap **16** is injected. By injecting flush water from the nozzle in the inner tank **14**, the flush water stored in the inner tank **14** can be entrained, and the flush water in the inner tank **14** can flow into the flush toilet main body by jet pump action. In addition to this jet pump (not shown), an arbitrary configuration that switches supply and stop of flush water to the flush toilet main body **2** may be used as the second water discharge/stop switching mechanism.

[0049] Furthermore, the ball tap **16**, which is the water supply valve, is configured so that flush water supplied from the water supply source **6** flows into the ball tap through an inflow pipe **16***a* and is configured to switch supply and stop of flush water to be stored in the outer tank **12** and the inner tank **14**.

[0050] Next, with reference to FIG. **4**, the configuration of the ball tap **16** will be described.

[0051] As shown in FIG. **4**, the ball tap **16** includes a main body **20** to which the inflow pipe **16***a* and an outflow pipe **16***b* are connected, a main valve body **20***a* disposed in the main body **20**, a valve seat **20***b* on which the main valve body **20***a* is seated, an arm **24** rotated by a float **22**, and a pilot valve **26** moved by the rotation of the arm **24**. That is, the ball tap **16** includes the float **22** that operates in conjunction with a water level in the flush water tank device **4**, and the float **22** is configured to supply flush water to the hydraulic drive mechanism **18** when lowering to a predetermined position.

[0052] The main body **20** is a member having a lower part including a connecting portion to the inflow pipe **16***a* and having one side including a connecting portion to the outflow pipe **16***b*. Furthermore, the valve seat **20***b* is formed inside the main body **20**, and the valve seat **20***b* communicates with the outflow pipe **16***b* connected to the connecting portion. Furthermore, inside the main body **20**, the main valve body **20***a* is disposed to open and close the valve seat **20***b* and is configured so that when the valve is opened, tap water flowing from the inflow pipe **16***a* flows through the valve seat **20***b*, to flow out to the outflow pipe **16***b*. The outflow pipe **16***b* is then connected to the hydraulic drive mechanism **18**.

[0053] The main valve body **20***a* is a diaphragm valve body that is generally disc-shaped and is attached to the main body **20** so that the main valve body can be seated on and unseated from the valve seat **20***b*. Furthermore, a bleed hole **20***c* is provided in a peripheral edge portion of the main valve body **20***a*. Furthermore, in the main valve body **20***a*, a pressure chamber **20***d* is formed on a side of the main valve body **20***a* opposite to the valve seat **20***b* (left side in FIG. **4**). That is, the pressure chamber **20***d* is defined by an inner wall surface of the main body **20** and the main valve body **20***a*, and when the pressure in the pressure chamber **20***d* increases, the main valve body **20***a* is pressed against the valve seat **20***b* and seated on the valve seat **20***b* by this pressure.

[0054] Furthermore, a pressure passage **20***e* extends upward to communicate with the pressure chamber **20***d* provided in the main body **20**, and a pilot valve port **26***a* is provided at an upper end of the pressure passage **20***e*. The pilot valve port **26***a* is opened upward and is configured to be opened and closed by the pilot valve **26**.

[0055] On one hand, the float **22** is supported by the arm **24**, and the arm **24** is rotatably supported by a support shaft **24***a*. Furthermore, the pilot valve **26** is coupled to the arm **24**, and the pilot valve **26** is configured to move in an up-down direction together with the rotation of the arm **24**. In the present embodiment, the float **22** is disposed in the outer tank **12** and moved vertically depending on the water level in the outer tank **12**. Consequently, when the water level in the outer tank **12** is raised to or above a predetermined water level, the float **22** is pushed upward, and the pilot valve **26** is accordingly moved downward and seated on the pilot valve port **26***a*, to close the pilot valve port. On the other hand, when the flush water in the outer tank **12** is discharged to lower the water level, the float **22** moves downward, and the pilot valve **26** moves upward, to open the pilot valve port **26***a*. Consequently, when toilet flushing is on standby in a state in which the water level in the outer tank **12** is higher than the predetermined water level, the pilot valve port **26***a* of the main body **20** is closed.

[0056] Furthermore, tap water flowing through the inflow pipe **16***a* into the main body **20** flows into an annular space around the valve seat **20***b* and flows from here through the bleed hole **20***c* of the main valve body **20***a* into the pressure chamber **20***d*. In this state in which the pilot valve port **26***a* is closed by the pilot valve **26**, there is no path for tap water flowing from the bleed hole **20***c* into the pressure chamber **20***d* to flow out, and the pressure in the pressure chamber **20***d* rises. When the pressure in the pressure chamber **20***d* rises in this way, the main valve body **20***a* is pressed toward the valve seat **20***b* (rightward in FIG. **4**) by this pressure, and the valve seat **20***b* is closed by the main valve body **20***a*.

[0057] In contrast, when the first discharge valve **8** is opened by the flushing operation and the water level in the outer tank **12** becomes lower than the predetermined water level, the float **22** descends, and the pilot valve **26** moves upward, to open the pilot valve port **26***a*. When the pilot valve port **26***a* is opened, water in the pressure chamber **20***d* flows out of the pilot valve port **26***a* and the pressure in the pressure chamber **20***d* decreases. Consequently, the main valve body **20***a* is moved to be pulled away from the valve seat **20***b* (leftward in FIG. **4**), to open the valve seat **20***b*. Consequently, in this state in which the pilot valve port **26***a* is opened, the pressure in the pressure chamber **20***d* does not increase, so that the valve seat **20***b* is opened.

[0058] Next, with reference to FIG. **5**, the configuration of the hydraulic drive mechanism **18** will be described.

[0059] The hydraulic drive mechanism **18** is configured to drive the second discharge valve **10** using a water supply pressure of the flush water supplied from the water supply to the flush water tank device **4**. Specifically, the hydraulic drive mechanism **18** includes a cylinder **18***a* into which water supplied from the ball tap **16** flows, a piston **18***b* slidably disposed in the cylinder **18***a*, and a rod **28** that protrudes from a lower end of the cylinder **18***a* to drive the second discharge valve **10**. Furthermore, a spring **18***c* is disposed inside the cylinder **18***a*, to bias the piston **18***b* downward, and a packing is attached to the piston **18***b*, to acquire watertightness between an inner wall surface of the cylinder **18***a* and the piston **18***b*. In addition, a clutch mechanism **30** is provided at a lower end of the rod **28**, and the rod **28** and the valve shaft **10***a* of the second discharge valve **10** are coupled/decoupled by the clutch mechanism **30**.

[0060] The cylinder **18***a* is a cylindrical member, has an axis oriented in a vertical direction and can slidably receive the piston **18***b* inside. Furthermore, the outflow pipe **16***b* extending from the ball tap **16** is connected to a lower end of the cylinder **18***a*, and the flush water flowing out of the ball tap **16** flows into the cylinder **18***a*. Consequently, the piston **18***b* in the cylinder **18***a* is pushed up against a biasing force of the spring **18***c* by the water flowing into the cylinder **18***a*.

[0061] In addition, an outflow hole is provided in an upper end of the cylinder **18***a*, and a water supply pipe **32** is connected to the outflow hole. Therefore, when water flows into the cylinder **18***a* through the outflow pipe **16***b* connected to a lower part of the cylinder **18***a*, the piston **18***b* is pushed upward from the lower part of the cylinder **18***a*. Then, when the piston **18***b* is pushed up to a position above the outflow hole, the water flowing into the cylinder **18***a* flows out of the outflow hole to the water supply pipe **32**.

[0062] The water supply pipe **32** is a pipe extending downward from the outflow hole provided in the upper end of the cylinder **18***a*. The flush water flowing out of the cylinder **18***a* flows into the inner tank **14** from a water supply port **32***a* at a lower end of the water supply pipe **32**. The water supply port **32***a* at the lower end of the water supply pipe **32** is located below the water surface of the flush water in the inner tank **14** on standby. Therefore, the ball tap **16** is configured to cause flush water to flow into the inner tank **14** from the water supply port **32***a* provided below the water surface of the inner tank **14**. Thus, since flush water is supplied into the inner tank **14** from below the water surface of the stored flush water, ripples on the water surface of the flush water flowing into the inner tank **14** can be suppressed.

[0063] The rod **28** is a bar-shaped member connected to a lower surface of the piston **18***b* and extends to protrude downward from the cylinder **18***a* via a through hole formed on a bottom surface of the cylinder **18***a*. Furthermore, the valve shaft **10***a* of the second discharge valve **10** is connected to the lower end of the rod **28** via the clutch mechanism **30**, and the rod **28** couples the piston **18***b* and the second discharge valve **10**. Consequently, when water flows into the cylinder **18***a* to push up the piston **18***b*, the rod **28** connected to the piston **18***b* lifts the second discharge valve **10** upward, and the second discharge valve **10** is opened.

[0064] Furthermore, a gap is provided between the rod **28** protruding from the bottom of the cylinder **18***a* and the inner wall of the through hole of the cylinder **18***a*, and part of the water flowing into the cylinder **18***a* flows out of this gap. The water flowing out of the gap flows into the inner tank **14**. This gap is comparatively narrow and has a large flow channel resistance. Therefore, even if water flows out of the gap, the pressure in the cylinder **18***a* increases due to the water flowing through the outflow pipe **16***b* into the cylinder **18***a*, and the piston **18***b* is pushed up against the biasing force of the spring **18***c*.

[0065] Furthermore, the clutch mechanism **30** removably couples the rod **28** and the second discharge valve **10**. The clutch mechanism **30** is configured to disconnect the valve shaft **10***a* of the second discharge valve **10** from the rod **28**, when the second discharge valve **10** is lifted up together with the rod **28** along a predetermined distance. While the clutch mechanism **30** is disconnected, the second discharge valve

10 is no longer linked to the movement of the piston **18***b* and the rod **28**, and the second discharge valve **10** descends with decrease in water level in the inner tank **14**, to close the drain port **14***a* of the inner tank **14**.

[0066] Next, newly with reference to FIG. **6**, an operation of the flush toilet apparatus **1** according to the embodiment of the present invention will be described.

[0067] FIG. **6** is a time chart showing the operation of the flush toilet apparatus **1** according to the embodiment of the present invention, and shows a state of jet spouting, a state of rim spouting, and a state of the ball tap in order from the top.

[0068] First, in a standby state for toilet flushing, the first drain port **12***a* of the outer tank **12** and the drain port **14***a* of the inner tank **14** are closed by the first discharge valve **8** and the second discharge valve **10**, respectively. Furthermore, in the standby state, flush water is stored up to a predetermined initial water level in each of the outer tank **12** and the inner tank **14**. The initial water level of the outer tank **12** is set higher than the predetermined initial water level. Thus, the pilot valve port **26***a* of the main body **20** (FIG. **4**) of the ball tap **16** is closed, and the valve seat **20***b* is closed by the main valve body **20***a*. In the present embodiment, the initial water level of the inner tank **14** is at a position higher than the initial water level of the outer tank **12**, and flush water is stored almost to the upper end in the inner tank **14**.

[0069] Next, at time t**1** in FIG. **6**, when the user rotates and operates the lever handle **4***a* (FIG. **3**) of the flush water tank device **4** to perform toilet flushing, the bead chain **8***a* connected to this handle pulls up the first discharge valve **8**. Consequently, the first discharge valve **8** is pulled away from the first drain port **12***a*, and the first drain port **12***a* is opened. When the first drain port **12***a* is opened, the flush water stored inside the outer tank **12** (outside the inner tank **14**) flows from the first drain port **12***a* into the rim conduit **2***e* (FIG. **2**) and is spouted from the rim spout port **2***c*. By rim spouting from the rim spout port **2***c*, a circulating flow is formed on the waste receiving surface of the bowl **2***a*, and the waste receiving surface is flushed.

[0070] As the flush water is drained from the first drain port **12***a*, the water level in the outer tank **12** decreases. Consequently, the float **22** of the ball tap **16** lowers to open the pilot valve **26** (FIG. **4**). This decreases the pressure in the pressure chamber **20***d*, to open the main valve body **20***a*, and flush water is supplied through the outflow pipe **16***b* to the hydraulic drive mechanism **18**.

[0071] When flush water is supplied to the hydraulic drive mechanism **18**, the flush water flowing into the cylinder **18***a* (FIG. **5**) pushes up the piston **18***b* against the biasing force of the spring **18***c*. Consequently, the rod **28** coupled to the piston **18***b* pulls up the valve shaft **10***a* of the second discharge valve **10**, and the drain port **14***a* of the inner tank **14** is opened. That is, the second discharge valve **10** is driven by the water supply pressure of tap water supplied via the ball tap **16** and is opened.

[0072] At time t**2** in FIG. **6**, the drain port **14***a* is opened, and the flush water stored in the inner tank **14** flows through the drain port **14***a* and the second drain port **12***b* into the jet conduit **2***f* (FIG. **2**) and is spouted from the jet spout port **2***d*. The jet spouting from the jet spout port **2***d* fills the discharge trap conduit **2***b* with water and induces siphon action. Due to occurrence of the siphon action, retained water and waste in the bowl **2***a* are suctioned into the discharge trap conduit **2***b* and drained to the sewage pipe (not shown).

[0073] Here, in a state in which the drain port **14***a* is closed, the interior of the jet conduit **2***f* is filled with flush water to a level of pooled water surface of the bowl **2***a*, and no flush water is in a space above this pooled water surface. Therefore, in a state before the second discharge valve **10** is opened, no flush water is at an upstream end in the jet conduit **2***f*.

[0074] When the second discharge valve **10** is opened, the space in which there is no flush water in the jet conduit **2***f* is filled in a brief time, and the flush water can be spouted from the jet spout port **2***d* at an early stage. In particular, according to the present embodiment, the cross-sectional area of an upper part of the inner tank **14** is larger than the cross-sectional area of a lower part, and hence the inner tank **14** includes a large volume at a high position. Consequently, a large amount of flush water having a high hydraulic head pressure can flow into the jet conduit **2***f*, and the instantaneous flow rate of the flush water spouted from the jet spout port **2***d* can be increased.

[0075] When the second discharge valve **10** is pulled up to the predetermined height together with the piston **18***b* of the hydraulic drive mechanism **18**, the valve shaft **10***a* of the second discharge valve **10** is disconnected from the rod **28** by the clutch mechanism **30** (FIG. **5**). Consequently, the second discharge valve **10** descends toward the drain port **14***a* as the water level in the inner tank **14** decreases. Then, at time t**3** in FIG. **6**, the second discharge valve **10** is seated at the drain port **14***a*, and the drain port **14***a* is closed. This stops the jet spouting from the jet spout port **2***d*.

[0076] In a state in which the main valve body **20***a* of the ball tap **16** is opened, the flush water supplied from the water supply source **6** (water supply) is supplied to the hydraulic drive mechanism **18** via the ball tap **16**, and flows through the water supply pipe **32** (FIG. **5**) connected to the cylinder **18***a* into the inner tank **14**. Here, in a state in which the second discharge valve **10** is opened, the flow rate of the flush water flowing out of the drain port **14***a* of the inner tank **14** is larger than the flow rate of the flush water flowing through the water supply port **32***a* of the water supply pipe **32** into the inner tank **14**, and hence the water level in the inner tank **14** decreases. Then, after the jet spouting is stopped at time t**3** in FIG. **6** (after the second discharge valve **10** is closed), the flush water flowing in from the water supply pipe **32** raises the water level in the inner tank **14**.

[0077] On the other hand, since the first discharge valve **8** is still in an opened state, the flush water in the outer tank **12** flows out of the first drain port **12***a*, and the water level in the outer tank **12** continues to decrease. Then, at time t**4** in FIG. **6**, when the water level in the outer tank **12** decreases to a predetermined dead water level, the first discharge valve **8** is seated on the first drain port **12***a*, and the first discharge valve **8** is closed. Consequently, the rim spouting from the rim spout port **2***c* is stopped.

[0078] Furthermore, even after the first discharge valve **8** is closed, the main valve body **20***a* of the ball tap **16** is maintained in an opened state, and the flush water supplied from the water supply source **6** (water supply) therefore flows into the inner tank **14** from the water supply port **32***a* of the water supply pipe **32** via the ball tap **16** and the hydraulic drive mechanism **18**. At this time, since the water supply port **32***a* of the water supply pipe **32** is opened in the lower part of the inner tank **14**, the water supply port **32***a* is located under the water surface of the flush water stored in the inner tank **14**, and flush water flows from stored water into the inner tank **14**. As a result, when the flush water flows from the water supply port **32***a* to the inner tank **14**, splashing of flush water and ripples on the water surface in the inner tank **14** are suppressed.

[0079] When the water level in the inner tank **14** rises due to the inflow of flush water, at time t**5** in FIG. **6** at which the inner tank **14** is full, the flush water begins to overflow from the inner tank **14**. In the present embodiment, since the upper end of the inner tank **14** is formed at the same height along the entire perimeter, flush water overflows from the entire perimeter of the inner tank **14** to flow into the outer tank **12**. In the present embodiment, since the edge of the inner tank **14** over which the flush water flows is formed extremely long, the water level of the flush water flowing over the edge of the inner tank **14** can be kept low. That is, the flush water overflowing from the inner tank **14** flows over the edge of the inner tank **14** while forming a thin water film on the edge of the inner tank **14** and flows into the outer tank **12**. In other words, since the edge of the inner tank **14** is formed long, the flush water inflowing from the water supply port **32***a* at a predetermined flow rate only forms the thin water film on the edge and can flow out.

[0080] As flush water flows into the outer tank **12**, the water level in the outer tank **12** begins to rise. Next, the water level in the outer tank **12** rises to a predetermined water level, and the float **22** of the ball tap **16** rises, to close the pilot valve **26** (FIG. **4**). Thus, when the pilot valve **26** is closed, the flush water flowing into the pressure chamber **20***d* from the bleed hole **20***c* provided in the main valve body **20***a* of the ball tap **16** cannot flow out, and the pressure in the pressure chamber **20***d* increases. Then, at time t**6** in

[0081] FIG. **6**, the main valve body **20***a* is pressed by the pressure in the pressure chamber **20***d* and seated on the valve seat **20***b*, and the main valve body **20***a* is closed. This stops water supply from the water supply source **6** via the ball tap **16** to the hydraulic drive mechanism **18** and stops the supply of flush water into the inner tank **14**.

[0082] Upon stopping the water supply to the hydraulic drive mechanism **18**, the piston **18***b* (FIG. **5**) in the cylinder **18***a* pushed up by the water supply is pushed down by the biasing force of the spring **18***c*. The rod **28** attached to the piston **18***b* accordingly lowers. The rod **28** lowers to a predetermined position, and the rod **28** is recoupled to the valve shaft **10***a* of the second discharge valve **10** by the clutch mechanism **30**. As described above, toilet flushing is ended once, and the flush toilet apparatus **1** returns to the standby state for the toilet flushing.

[0083] According to the flush water tank device **4** of the embodiment of the present invention, the ball tap **16** is configured to supply flush water into the inner tank **14**, and the flush water supplied into the inner tank **14** overflows from the entire perimeter of the inner tank **14** to flow into the outer tank **12**. Therefore, the edge over which the flush water flows can be lengthened, and the water level of flush water flowing over the edge of the inner tank **14** can be kept low. Consequently, the space to be provided above the inner tank **14** can be set low, and the flush water tank device **4** can be designed with a low profile.

[0084] According to the flush water tank device **4** of the present embodiment, since the length of the outer perimeter of the upper end of the inner tank **14** is set to ½ or more of the length of the outer perimeter of the upper end of the outer tank **12**, the edge of the inner tank **14** over which the flush water flows can be set sufficiently long, and the water level of the flush water flowing over the edge of the inner tank **14** can be kept low.

[0085] According to the flush water tank device **4** of the present embodiment, since the inner tank **14** is configured so that the cross-sectional area of the upper part of the inner tank **14** is larger than the cross-sectional area of the lower part, the edge of the inner tank over which the flush water flows can be formed long. As a result, the water level of the flush water flowing over the edge of the inner tank **14** can be kept low. Furthermore, since the inner tank **14** is configured so that the cross-sectional area of the upper part is larger than the cross-sectional area of the lower part, a large amount of flush water having a large head pressure is stored in the inner tank **14**. Consequently, the instantaneous flow rate of flush water drained from the inner tank **14** can be increased, and the flush toilet main body **2** can be effectively flushed.

[0086] According to the flush water tank device **4** of the present embodiment, since the ball tap **16** is configured to cause flush water to flow into the inner tank **14** from the water supply port **32***a* provided below the water surface of the inner tank **14**, rippling of the flush water flowing into the inner tank **14** from the water supply port **32***a* can be suppressed. As a result, even if the space above the inner tank **14** is set low, concern of equipment disposed above the inner tank **14** being covered with water can be reduced, and the flush water tank device **4** can be designed with the low profile.

[0087] According to the flush water tank device **4** of the present embodiment, since the inner tank **14** is configured so that the length of the outer perimeter of the upper end of the inner tank is longer than the length of the outer perimeter of the upper end of the outer tank **12**, the edge of the inner tank **14** over which flush water flows can be set extremely long, and the water level of the flush water flowing over the edge of the inner tank **14** can be kept sufficiently low.

[0088] As described above, the flush water tank device of the embodiment of the present invention and the flush toilet apparatus including the same have been described, and various changes can be made to the above-described embodiment. In particular, in the above-described embodiment, the flush water stored in the inner tank is spouted from the jet spout port, and the flush water stored in the outer tank is spouted from the rim spout port. In contrast, as a modification, the present invention may be configured so that the flush water of the inner tank is spouted from the rim spout port, and the flush water of the outer tank is spouted from the jet spout port. Alternatively, the present invention can be applied to a flush water tank device that supplies flush water to a flush toilet main body including a spout port other than the rim spout port and the jet spout port.

[0089] Furthermore, in the above-described embodiment, the ball tap is used as the water supply valve for switching the supply and stop of flush water to be stored in the outer tank and the inner tank, and a water supply valve other than the ball tap may be used. Furthermore, in the above-described embodiment, the second discharge valve is driven by the hydraulic drive mechanism, and the hydraulic drive mechanism may not be provided.

REFERENCE SIGNS LIST

[0090] **1** flush toilet apparatus

[0091] **2** flush toilet main body

[0092] **2***a* bowl

[0093] **2***b* discharge trap conduit
[0094] **2***c* rim spout port
[0095] **2***d* jet spout port
[0096] **2***e* rim conduit
[0097] **2***f* jet conduit
[0098] **4** flush water tank device
[0099] **4***a* lever handle
[0100] **6** water supply source
[0101] **8** first discharge valve (first water discharge/stop switching mechanism)
[0102] **8***a* bead chain
[0103] **10** second discharge valve (second water discharge/stop switching mechanism)
[0104] **10***a* valve shaft
[0105] **12** outer tank
[0106] **12***a* first drain port
[0107] **12***b* second drain port
[0108] **14** inner tank
[0109] **14***a* drain port
[0110] **14***b* drain port forming member
[0111] **14***c* cutout
[0112] **14***d* cutout
[0113] **16** ball tap (water supply valve)
[0114] **16***a* inflow pipe
[0115] **16***b* outflow pipe
[0116] **18** hydraulic drive mechanism
[0117] **18***a* cylinder
[0118] **18***b* piston
[0119] **18***c* spring
[0120] **20** main body
[0121] **20***a* main valve body
[0122] **20***b* valve seat
[0123] **20***c* bleed hole
[0124] **20***d* pressure chamber
[0125] **20***e* pressure passage
[0126] **22** float
[0127] **24** arm
[0128] **24***a* support shaft
[0129] **26** pilot valve
[0130] **26***a* pilot valve port
[0131] **28** rod
[0132] **30** clutch mechanism
[0133] **32** water supply pipe
[0134] **32** water supply port

1. A flush water tank device that is attached to a flush toilet main body for use to supply flush water, comprising:
an outer tank that stores flush water to be supplied to the flush toilet main body,
an inner tank made of resin, that is disposed inside the outer tank and stores flush water to be supplied to the flush toilet main body,
a water supply valve that switches supply and stop of flush water to be stored in the outer tank and the inner tank,
a first water discharge/stop switching mechanism disposed inside the outer tank and outside the inner tank to switch supply and stop of flush water stored in the outer tank to the flush toilet main body, and
a second water discharge/stop switching mechanism disposed inside the inner tank to switch supply and stop of flush water stored in the inner tank to the flush toilet main body, wherein
the water supply valve is configured to supply flush water into the inner tank, and the flush water supplied into the inner tank overflows from two or more sides of the inner tank to flow into the outer tank.

2. The flush water tank device according to claim **1**, wherein the inner tank is configured so that a length of an outer perimeter of an upper end of the inner tank is ½ or more of a length of an outer perimeter of an upper end of the outer tank.

3. The flush water tank device according to claim **1**, wherein the inner tank is configured so that a cross-sectional area varies in a height direction and so that a cross-sectional area of an upper part of the inner tank is larger than a cross-sectional area of a lower part.

4. The flush water tank device according to claim **1**, wherein the water supply valve is configured to cause flush water to flow into the inner tank from a water supply port provided below the water surface of the inner tank.

5. The flush water tank device according to claim **1**, wherein the inner tank is configured so that a length of an outer perimeter of an upper end of the inner tank is longer than a length of an outer perimeter of an upper end of the outer tank.

6. A flush toilet apparatus which is flushed with flush water stored in a flush water tank, comprising:
a flush toilet main body including a bowl, a rim spout port, and a jet spout port, and
the flush water tank device according to claim **1** that supplies flush water to the rim spout port and the jet spout port of the flush toilet main body.

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